

Mechanistic insight into the catalytically active phase of CO₂ hydrogenation on Cu/ZnO catalyst

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ARTICLE INFO

Keywords:

CO₂ hydrogenation
Methanol synthesis
ZnOH/Cu active phase

ABSTRACT

The catalytical active phase and reaction mechanism of methanol synthesis by hydrogenation of carbon dioxide on the surface of Cu/ZnO/Al₂O₃ catalyst become controversial topics in recent years. In this work, density functional theory calculations are employed to explore the possible mechanisms of the CO₂ hydrogenation on three possible active phases, namely Zn/Cu interface, ZnO/Cu interface and hydroxylated Zn/Cu surface (i.e., ZnOH/Cu), which may exist in Cu/ZnO/Al₂O₃ catalyst under realistic conditions. It is more favorable that CO₂ activation takes place by the formate pathway rather than the Reverse Water-Gas Shift pathway. Furthermore, the surface hydroxyl group could significantly increase the activity of formate pathway by improving the hydrogenation steps of HCOO to HCOOH and H₃CO to H₃COH, which are usually deemed as the rate-determining steps. These results suggest that the ZnOH/Cu phase is not only the most stable but also the most active phase for CO₂ activation among Zn/Cu, ZnO/Cu and ZnOH/Cu, which could provide more insight into the design of highly efficient catalyst by tuning the surface active phase under operative conditions.

1. Introduction

The industrial conversion of carbon dioxide could not only reduce the dependence on traditional fossil fuels but also mitigate the greenhouse effect to improve the environment [1–4]. Methanol, as one of the important product of CO₂ hydrogenation, is one of the necessary organic chemical raw materials in industry, which can also be used as the clean energy source in fuel cells [5]. When compared to other products of CO₂ hydrogenation such as formic acid and methane, methanol has higher calorific value and higher equivalent hydrogen storage capacity. Because of its broad industrial application prospects, the methanol synthesis by CO₂ hydrogenation has been widely studied in recent years [6–21].

The synthesis of methanol in industry mainly takes place by the reaction of CO₂, CO, and H₂ gas, where the copper-based catalyst [22] Cu/ZnO/Al₂O₃ is typically applied under high temperature (450–600 K) and high pressure (50–100 atm). Despite extensive industrialization of methanol synthesis from carbon dioxide mixed gas,

there still exist some debates on the reaction mechanism and active sites because of the surface complexity of the catalyst and the harsh reaction conditions. Both carbon dioxide and carbon monoxide proved to be the carbon sources in methanol synthesis, depending on different reaction conditions and catalyst compositions [23–27].

For the methanol synthesis there are usually two traditional reaction pathways [8,28–30]: one is the formate pathway and the other is the Reverse Water-Gas Shift Reaction pathway followed by CO hydrogenation (RWGS + CO-Hydro), as shown in Fig. 1. In the formate pathway, carbon dioxide is first hydrogenated to HCOO species, and then to HCOOH and H₂COOH species. Afterwards H₂COOH will break into H₂CO and OH species. Finally, the H₂CO species will be gradually hydrogenated to H₃COH. In RWGS + CO-Hydro pathway, the carbon dioxide is first hydrogenated to COOH, and then it will break down to produce CO followed by hydrogenation to H₃COH step by step [6].

It was reported that the reaction over Cu-based catalysts where CO₂ hydrogenates to methanol shows strong sensitivity to their microstructures [31–35]. The size of copper particle is crucial to the reaction

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<https://doi.org/10.1016/j.apsusc.2020.146481>

Received 20 November 2019; Received in revised form 24 March 2020; Accepted 7 April 2020

Available online 29 April 2020

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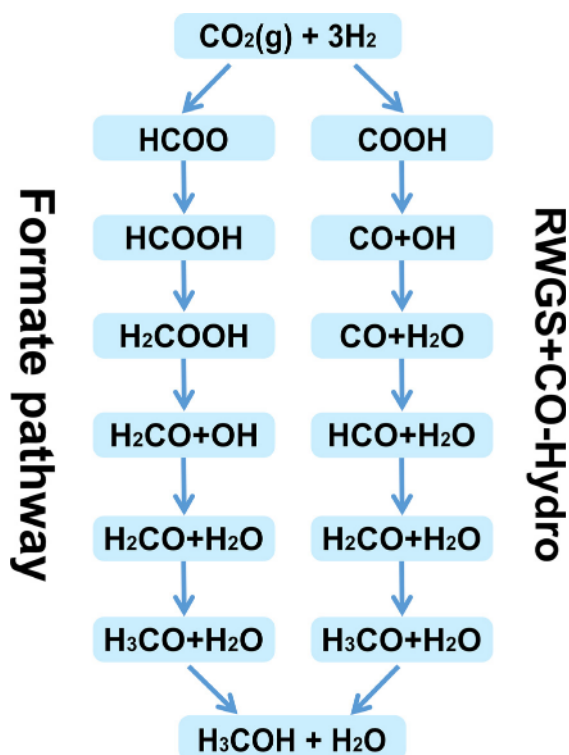


Fig. 1. Succinct reaction network of the formate pathway and the RWGS + CO-Hydro pathway.

activity, and there is a linear relevance between the methanol synthesis activity of Cu/ZnO/Al₂O₃ catalysts and their total copper surface area [25,36,37]. In addition, zinc or zinc oxide usually plays an essential role in the hydrogenation of carbon dioxide [38–40], where the presence of zinc promotes the activity of copper-based catalysts [19]. Many studies have reported that there is a synergistic effect between zinc (or zinc oxide) and copper [7,10,15,41–45]. According to Behrens et al. [15], the activity of methanol synthesis is directly related to the concentration of the stacking dislocation defect sites on Cu nanoparticles, and the Zn^{δ+} species near the Cu defect site could accelerate the reaction by the dynamic strong metal-support interaction (SMSI). Shik and co-workers [35] found that the surface morphology of zinc oxide and its interaction with copper have a significant influence on methanol synthesis. Moreover, the effects of supports have also been extensively studied [46,47]. Through experimental observation and theoretical calculation, Larmier and co-workers [48] highlighted the important molecular interaction of interface between the copper particles and zirconia supports.

The stability of intermediate species is of vital importance to the activity and selectivity in methanol synthesis [49,50]. Experimental characterization has shown that the formate species can accumulate on the surface of catalyst poisoning the catalyst [33]. Theoretical calculations verify that the barrier for converting HCOO to HCOOH is very high due to the strong binding of HCOO species on zinc. As a result, the conversion of the formate species is often considered as a rate-determining step during the overall synthesis of methanol.

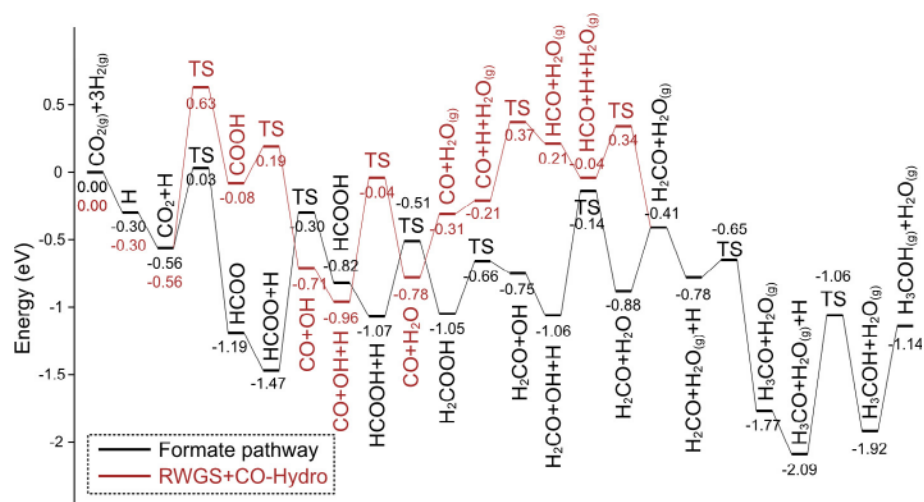
Besides those insights into the mechanism, the active site of copper-based catalysts is still in heated debate. Recently, Kettle and co-workers suggest the ZnO-Cu interface as the active site for methanol synthesis [18], which is contrary to the traditional point of view supporting the non-oxidized Zn/Cu [45,51]. According to a series of characterization, Kettle [18] and co-workers has proven that the surface zinc-copper alloy is oxidized to zinc oxide-copper in experiment, which significantly promotes methanol synthesis. Their theoretical calculation also shows the hydrogenation of zinc oxide is difficult. Hence the surface of copper-zinc alloy would eventually be oxidized and stable in the form of

copper-zinc oxide. However, Nakamura and co-workers [51] argued this view was not rigorous, because evacuation and temperature is crucial for the Zn state throughout the experiment. Oxygen may be formed by decomposition of formate intermediates under XPS evacuation condition, which in the following process oxidizes Zn. Till now, there is still no the decisive experimental evidence of the active center, which is mainly attributed to the complex structure of Cu/ZnO/Al₂O₃ catalyst and the difficulty on identification of Zn valence state. Whereas in this peculiar circumstance, the theoretical calculation can provide a better understanding of the mechanism to explore the activity phase.

In this work density functional theory (DFT) calculations are employed to reveal the reaction mechanism of methanol synthesis on the zinc-copper alloy. Our results show that the surface of copper-zinc alloy can be hydroxylated in the reaction process. Comparing to the Zn/Cu and ZnO/Cu interface, ZnOH/Cu is thermodynamically more stable. Furthermore, a systematic comparison on the reaction pathways of methanol synthesis is manipulated among the Zn/Cu, ZnO/Cu, and ZnOH/Cu model, in which the result does show the highest catalytic activity on ZnOH/Cu active phase. Especially in the hydrogenation steps of HCOO to HCOOH and H₂CO to H₃COH, which are usually deemed as the rate-determining steps and reported by both experimental and theoretical studies [18,33], on the ZnOH/Cu surface the barriers dramatically reduced. All these computational results show the ZnOH/Cu phase is the most active phase for CO₂ activation among Zn/Cu, ZnO/Cu and ZnOH/Cu.

2. Computational details

Spin-polarized density functional theory (DFT) calculations were carried out by using the Vienna ab-initio simulation package (VASP) code [52–54]. The projector augmented wave (PAW) method with a kinetic cutoff energy of 400 eV was employed [55]. Experimental observation shows both Cu (1 1 1) and Cu (2 1 1) surfaces exist on the Cu catalyst [15]. Although according to our calculation in Fig. S1, the surface energy of Cu step (2 1 1) is slightly higher than that of Cu flat (1 1 1), it is observed in experiment and theoretical calculation that the catalytic activity of Cu (2 1 1) surface is better than Cu (1 1 1) surface [15,34,43]. Therefore, the Cu (2 1 1) surface was employed here to study the mechanism of CO₂ hydrogenation to methanol and modeled by a five-layer 2 × 3 unit cell, in which copper atoms at the step edges were replaced by zinc. The model of copper-zinc oxide was established on the base of copper-zinc alloy, where oxygen atoms were embedded in the step of the lattice to form ZnO/Cu, and the ZnOH/Cu was similarly built by including the hydroxyls. The 3D configurations of all these models are in detail shown in Fig. S2. During the geometry optimization, the bottom two layers of copper surface were fixed while the others were allowed to relax. The vacuum spacing of 15 Å was added to avoid interaction between repeated slabs. Gaussian smearing method with an electronic temperature of k_BT = 0.05 eV was used to achieve rapid convergence. The van der Waals force was taken into account by involving D3 corrections [56,57]. The integration on Brillouin-zone was sampled with 3 × 3 × 1 Monkhorst-Pack grid. The iteration convergence of atomic force was set to be 0.02 eV/Å in geometrical optimization and 0.05 eV/Å in transition state calculation. The CI-NEB [58] method was first used to search the transition state. When sometimes it was hard to converge the atomic force to 0.05 eV/Å, further dimer [59] calculations were employed with the estimated TS configuration obtained by CI-NEB making the atomic force converged to target criterion. The adsorption energies were defined as E_{ad} = E_{total} - E_i - E_{slab}, where E_{total}, E_i, E_{slab} represented the total energy of the slab with the adsorbed species, the energy of the adsorbed species, and the energy of the clean slab, respectively. In addition, the dipole correction of the polarization effect on adsorption energy was tested and shown to be negligible in Table S1, thus was not involved in the adsorption energies.



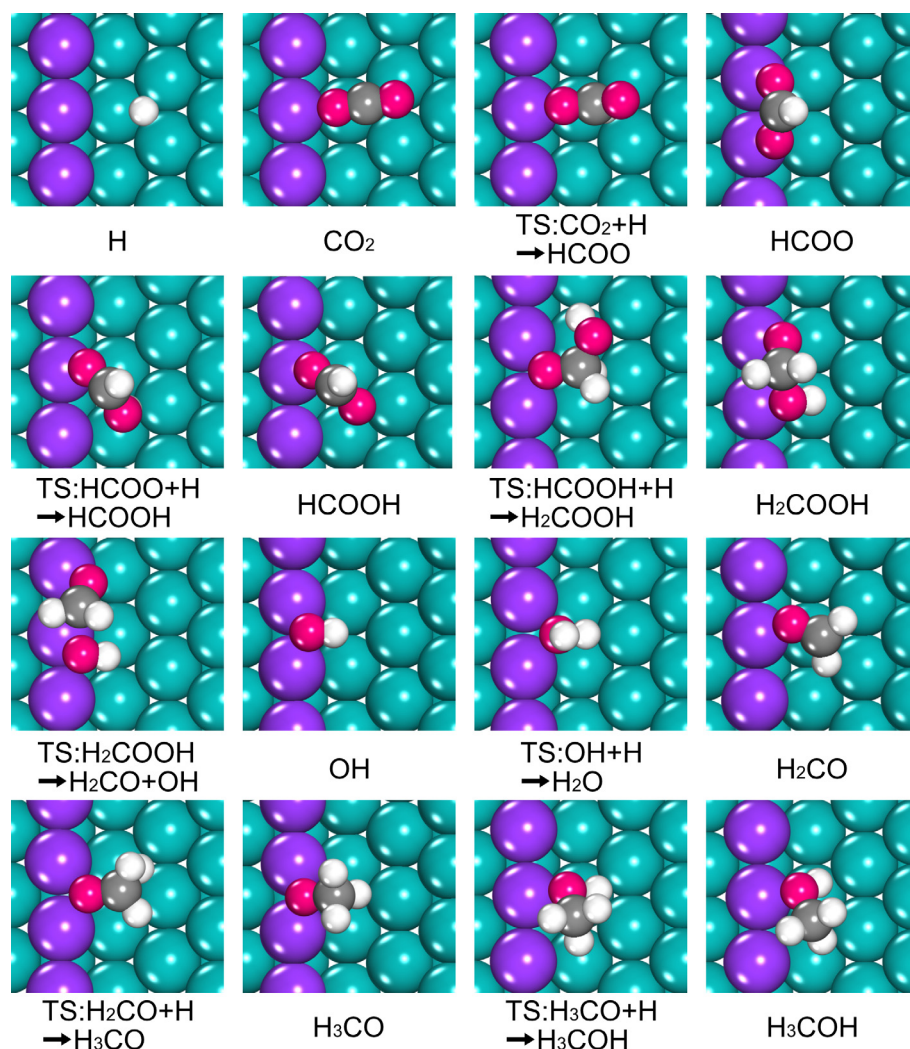


Fig. 3. The 3D configurations of the important intermediates and transition states along the formate pathway. The adsorbed carbon dioxide is in its physical adsorption state. Cu, Zn, C, O and H are represented by cyan, purple, red, grey and white color respectively.

ZnO layer making the ZnO/Cu interface as the multiple active sites for CO₂ hydrogenation.

However, soon afterwards this standpoint was argued by Nakamura and Fujitani [51] with two queries raised. First the condition in XPS detection is different from that in real catalytic reaction. Under vacuum in XPS, the detected oxygen can stem from the decomposition of intermediates rather than the ever-presented ZnO during reaction. Moreover, the specific investigations about the Zn L₃M_{4,5}M_{4,5} Auger lines, O1s lines and the C1s lines were not performed in Kattel's result [18]. All these can result in an unreasonable judgement on the attribution of active sites. In fact, the core problem whether the activate site is Zn/Cu or ZnO/Cu interface, quite depends on whether O can be stable on the surface of zinc-copper alloy in the real catalytic condition of high temperature and pressure. However, due to the limitation of experimental characterization, it is hard to detect all the surface species of catalyst in real reaction process. Hence, besides the model of Zn/Cu, the catalytic reactions on ZnO/Cu and ZnOH/Cu surfaces are also considered in the following DFT calculation.

The ZnO/Cu model is considered in the present work, where the surface oxygen can come from the decomposition of carbon dioxide at the Zn/Cu (2 1 1) step sites. The calculated barriers and reaction energies of this decomposition step are 0.60 and 0.08 eV respectively, as shown in Table 1 and Fig. S3. Similar to the situation on clean Zn/Cu surface, the formate pathway is also more favorable here comparing to the RWGS + CO-hydro pathway, due to a higher barrier of CO₂

hydrogenation to COOH ($E_a = 1.24$ eV).

Besides the debated active sites of ZnO/Cu and Zn/Cu, considering the reductive atmosphere of the overflowing hydrogen gas in experiment, the new active phase as ZnOH/Cu could be formed during the methanol synthesis. In our calculation, it is founded that the splitting of H₂COOH can lead to the formation of hydroxyl group (OH) at Zn site, forming ZnOH/Cu active phase where this OH will bond tightly with zinc atom with the adsorption energy up to 4.12 eV (Table S3). Due to the complexity of Zn/Cu (2 1 1) step surface, further tests on different configurations of intermediates are employed. It was found that H₂COOH prefers to lie on the Zn bridge site, where both O atoms are bonded with their neighboring Zn, followed by the splitting of H₂COOH and formation of ZnOH, as shown Fig. 3. In addition, the ZnO/Cu phase itself is not stable, which can be hydrogenated to OH group by the hydrogen atom transfer from the Cu site with the considerably small barrier of 0.60 eV. What is more, the generated water molecule as the CO₂ hydrogenation product can easily promote the formation of surface hydroxyls [8,63], where the adsorbed O could react with water generating two hydroxyls with a very low barrier of 0.09 eV. As the reaction proceeds, the surface of copper-zinc alloy will be gradually hydroxylated. Therefore, CO₂ hydrogenation reaction should also be considered on the ZnOH/Cu active phase. As shown in Fig. 5, the corresponding intermediates and transition states on the surface of ZnOH/Cu are similar to those on Zn/Cu clean surface with the relative energies and barriers of each steps listed in Table 1.

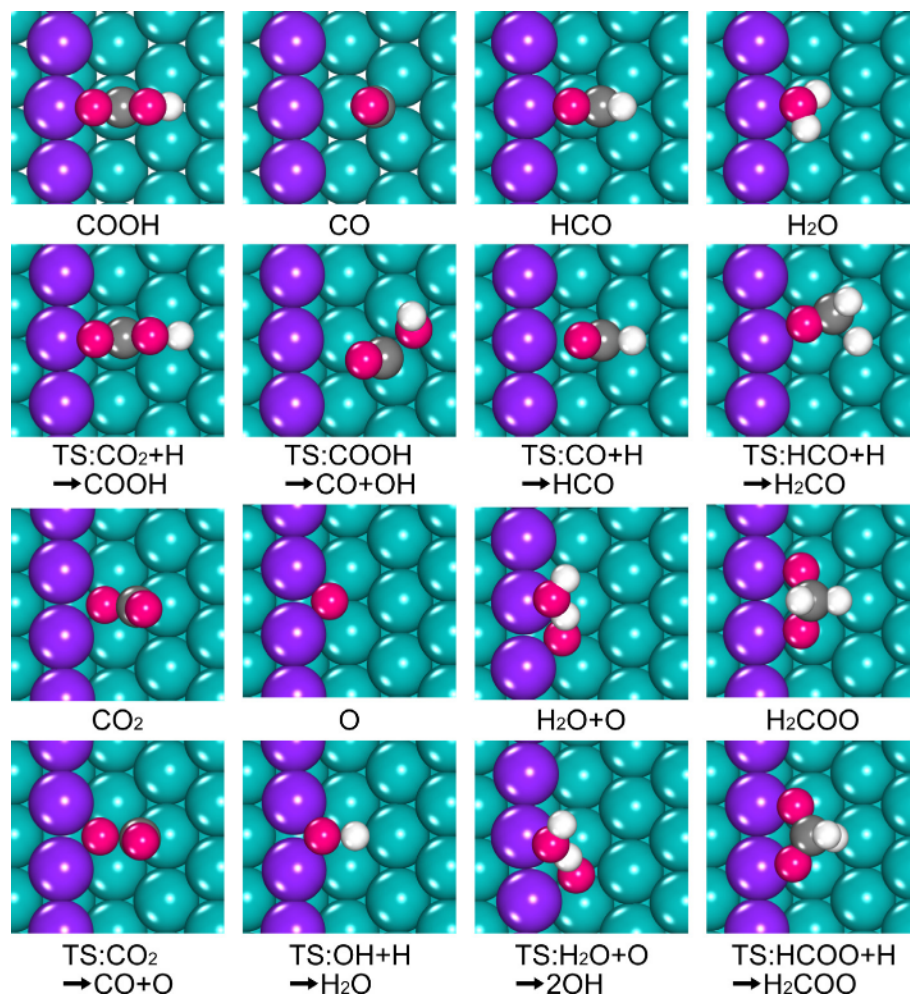


Fig. 4. The 3D configurations of the important intermediates and transition states along the RWGS + CO-hydro pathway. The adsorbed water is in its physical adsorption state, whereas the adsorbed carbon dioxide is in its chemical adsorption state preparing to dissociate on the Zn/Cu step sites. Colors are same in Fig. 3.

Table 1

The barriers and reaction energies of the elementary steps on the Zn/Cu (2 1 1), ZnO/Cu (2 1 1) and ZnOH/Cu (2 1 1) surface. The adsorbed carbon dioxide and water are both physical adsorption state.

Elementary Step	Zn/Cu		ZnO/Cu		ZnOH/Cu	
	E_a (eV)	ΔE (eV)	E_a (eV)	ΔE (eV)	E_a (eV)	ΔE (eV)
$\text{CO}_2 + \text{H} \rightarrow \text{HCOO}$	0.59	-0.63	0.55	-0.57	0.56	-0.28
$\text{HCOO} + \text{H} \rightarrow \text{HCOOH}$	1.17	0.65	1.09	0.6	0.75	0.3
$\text{HCOOH} + \text{H} \rightarrow \text{H}_2\text{COOH}$	0.56	0.02	0.46	-0.42	0.68	0.05
$\text{H}_2\text{COOH} \rightarrow \text{H}_2\text{CO} + \text{OH}$	0.39	0.3	1.07	0.93	0.68	0.54
$\text{OH} + \text{H} \rightarrow \text{H}_2\text{O}$	0.92	0.18	0.74	0.16	0.74	0.16
$\text{H}_2\text{CO} + \text{H} \rightarrow \text{H}_3\text{CO}$	0.13	-0.99	0.02	-1.05	0.21	-0.86
$\text{H}_3\text{CO} + \text{H} \rightarrow \text{H}_3\text{COH}$	1.03	0.16	1.02	0.19	0.71	0
$\text{H}_2\text{O} + \text{O} \rightarrow 2\text{OH}$	0.09	-0.49	-	-	-	-
$\text{O} + \text{H} \rightarrow \text{OH}$	0.6	-0.72	-	-	-	-
$\text{CO}_2 \rightarrow \text{O} + \text{CO}$	0.60	0.08	-	-	-	-

3.3. Comparison on the mechanisms among Zn/Cu, ZnO/Cu and ZnOH/Cu.

In this section, a systematic comparison on the mechanisms among the Zn/Cu alloy, ZnO/Cu, and ZnOH/Cu surfaces is made to further explore the most active phase in methanol synthesis. As shown in Fig. 6, the hydrogenation of HCOO to HCOOH and H_3CO to H_3COH are the rate-determining steps among the multiple elementary steps in methanol synthesis on not only the Zn/Cu surface but also the ZnO/Cu and

ZnOH/Cu surfaces. In these two steps, the ZnOH/Cu exhibits the lowest barrier among Zn/Cu alloy, ZnO/Cu, and ZnOH/Cu, and when comparing the overall energy diagrams the ZnOH/Cu is also the most active phase. In addition, the zero-point energy correction (ZPC), entropy contribution and free energy values are also estimated at the typical experimental temperature (550 K) [15,18], as listed in detail in Table S4, with the free energy diagram of the reaction pathways on Zn/Cu ZnO/Cu and ZnOH/Cu surfaces shown in Fig. S4. In accordance with the relative energy diagram in Fig. 6, the free energy values also support the ZnOH/Cu is the most active phase.

Generally speaking, due to the reducibility of Zn and the strong Zn-O bond, the hydrogenation on oxygen atom of the various intermediates is relatively difficult comparing to that on carbon atom. When the surface of Zn/Cu alloy is oxidized to ZnO/Cu or hydroxylated to ZnOH/Cu, the already bonded O or OH reduces the O affinity of Zn, leading to the obvious promoting effect on the two rate-determining steps. What should be noticed is that, as shown in Table 1 and Fig. 6, along the formate pathway not all reaction steps exhibit the lower energy barriers on ZnOH/Cu and ZnO/Cu surface comparing to those on Zn/Cu. For example, the splitting of H_2COOH is more difficult on the surface of ZnO/Cu due to the strong hydrogen bonding between H_2COOH and ZnO ($\text{H}_2\text{COOH-OZn}$), where extra energy is needed to break this hydrogen bond. The activation energies of this step on ZnOH/Cu ($E_a = 0.68$ eV) and ZnO/Cu ($E_a = 1.07$ eV) are both higher than that on Zn/Cu alloy ($E_a = 0.39$ eV). Moreover, the hydrogenation of H_2CO to H_3CO is an extremely easy step on the ZnO/Cu with only the barrier of 0.02 eV, while that barriers on the surface of Zn/Cu and ZnOH/Cu

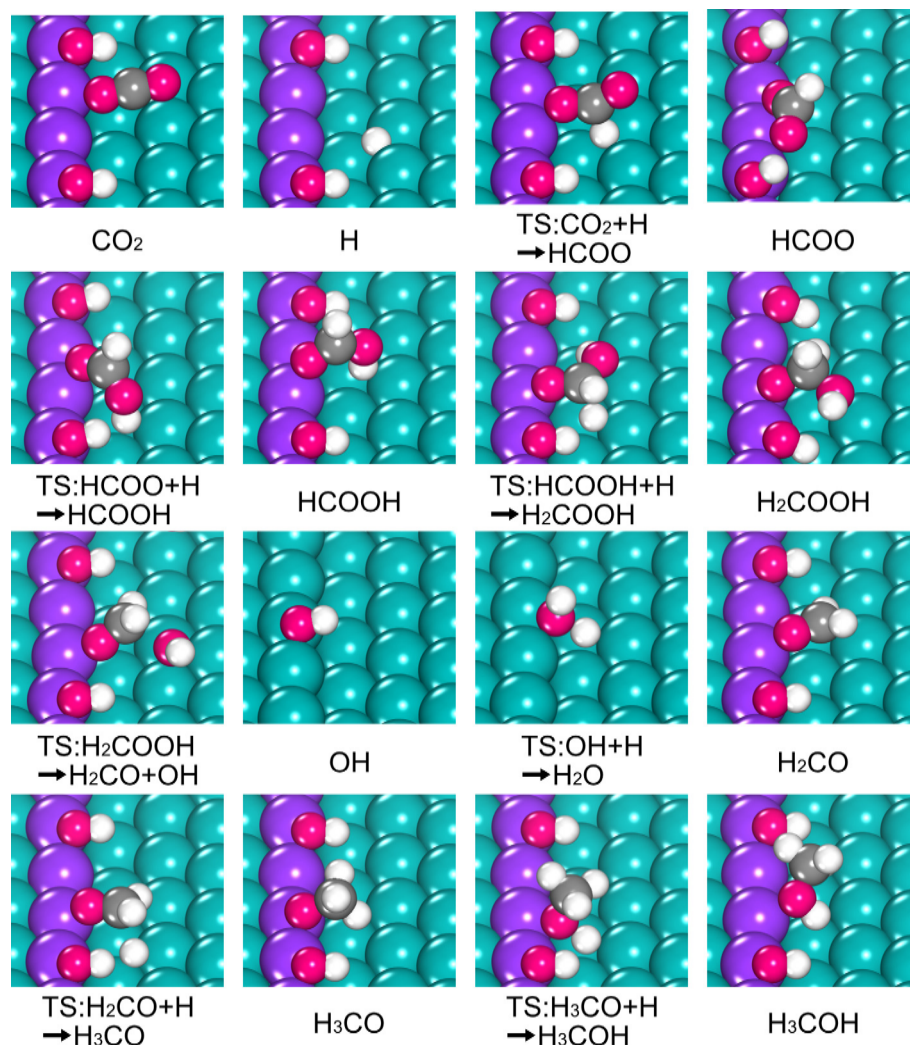


Fig. 5. The 3D configurations of the important intermediates and transition states on the surface of Cu/ZnOH. Colors are the same in Fig. 3.

becomes slightly higher to be 0.13 eV and 0.21 eV, respectively.

To investigate the interaction between the key adsorption species and the active phase, the charge density distribution analysis of HCOO adsorbed on the surfaces of Zn/Cu, ZnO/Cu and ZnOH/Cu is implemented. The charge density difference is defined as $\Delta\rho = \rho_{\text{HCOO} + \text{slab}} - \rho_{\text{slab}} - \rho_{\text{HCOO}}$, where, ρ_{HCOO} and ρ_{slab} represents the charge

density of HCOO and the surfaces (i.e., Zn/Cu, ZnO/Cu and ZnOH/Cu) respectively, and $\rho_{\text{HCOO} + \text{slab}}$ represents the charge density of HCOO adsorbed on the corresponding surface. In Fig. 7 yellow and blue iso-surface respectively represents the accumulation and depletion of electrons. According to the Bader charge analysis for the periodic DFT, the zinc can transfer only a small amount of electrons to copper ($0.1|e^-|$)

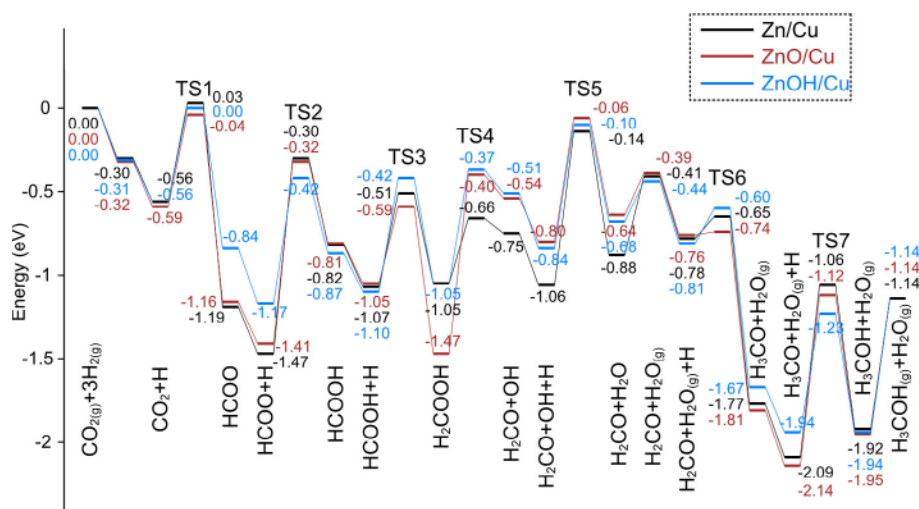


Fig. 6. Relative energies along the reaction pathway on different surfaces (i.e., Zn/Cu, ZnO/Cu and ZnOH/Cu). For concise, only the favorable formate pathways are shown. The black, red, and blue lines represent the pathways on Zn/Cu, ZnO/Cu and ZnOH/Cu surfaces, respectively. The corresponding free energies are estimated and shown in Fig. S4.

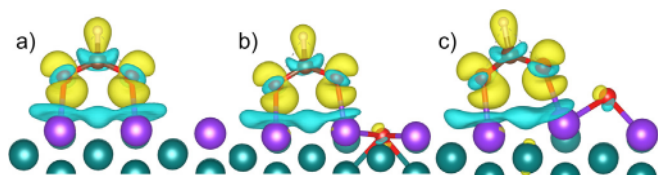


Fig. 7. Charge density distribution of HCOO adsorbed on the surface of Zn/Cu (2 1 1), ZnO/Cu (2 1 1), and ZnOH/Cu (2 1 1). The yellow and blue contours represent charge accumulation and charge depletion. The isosurfaces are plotted at the same scale of $\pm 0.02 |e| \text{ \AA}^{-3}$.

but considerably some amount of electrons to hydroxyl groups ($1.6|e|$), leading to the weaker O-Zn bond on the ZnOH/Cu surface, which could directly influence the adsorption energy of HCOO intermediate. The adsorption energy of HCOO on Zn/Cu (-3.69 eV) is very close to that on ZnO/Cu (-3.66 eV), but that energy on the ZnOH/Cu (-3.34 eV) is considerably smaller (Figs. S3 and Table S3). This trend is also valid on the Zn-O bond length. In the HCOO adsorption structure, the O-Zn bonds on Zn/Cu ($2.01 \text{ \AA}$) and on ZnO/Cu ($2.03 \text{ \AA}$) become elongated to $2.13 \text{ \AA}$ on ZnOH/Cu. In addition, because of the hydrogen bond between the hydroxyl groups on ZnOH/Cu surface and the oxygen atoms of intermediates, the hydroxyl group itself may act as a proton transfer agent for the process of H_3CO hydrogenation ($\text{H}_3\text{CO} + \text{OH} = \text{H}_3\text{COH} + \text{O}$). With the barrier of 0.96 eV , it is less favorable than direct hydrogenation, which contributes to a minor pathway.

Finally, two key steps (the hydrogenation of HCOO to HCOOH and H_3CO to H_3COH) are taken out separately for the further intuitive comparison among Zn/Cu, ZnO/Cu and ZnOH/Cu surfaces. It can be seen in Fig. 8 that the barrier for the hydrogenation of HCOO to HCOOH is 1.08 eV on the ZnO/Cu surface, which is only slightly lower compared with Zn/Cu surface, but the ZnOH/Cu surface shows a more obvious reaction promotion effect with the barrier of only 0.75 eV as the most active phase. Such dramatical promotion effect of ZnOH/Cu also takes place in the step of the H_3CO hydrogenation to H_3COH , in which the barrier on ZnOH/Cu is lower than those on both Zn/Cu and ZnO/Cu by around 0.3 eV . In addition, by employing the free energy barriers, the corresponding estimated rate constants of the two key steps could be obtained, as shown in Table S5. The ZnOH/Cu phase shows much higher reaction rate in the hydrogenation step of both HCOO and H_3CO species comparing to those on Zn/Cu and ZnO/Cu, even by 2–3 orders, which vividly confirms our conclusion that ZnOH/Cu is the most active phase in the Zn/Cu catalytic system of CO_2 hydrogenation to methanol.

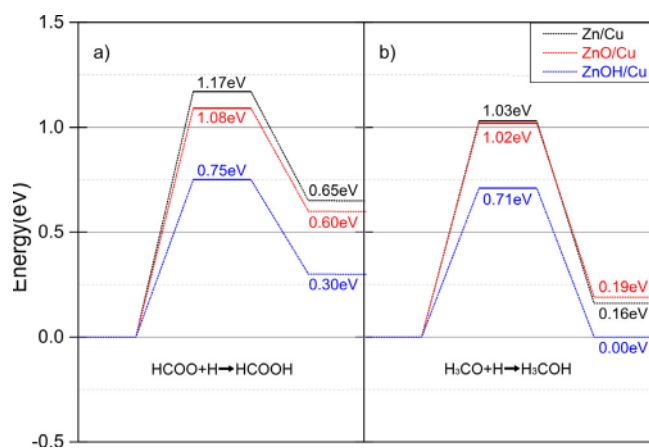


Fig. 8. The comparison among Zn/Cu, ZnO/Cu, and ZnOH/Cu surfaces on the barriers and reaction energies of the two key steps in methanol synthesis. Left a): the hydrogenation of HCOO to HCOOH; right b): the hydrogenation of H_3CO to H_3COH .

What should be kept in mind is that the concentration of OH group on the Zn/Cu surface should not be too large. Otherwise, they could squeeze the reaction sites and may poison the catalyst. Therefore, it is very important to control the proportion of mixed gases to provide an appropriate reductive atmosphere in methanol synthesis. Consequently, the coverage ratio of OH group on the Zn/Cu alloy should be controlled within a suitable range to promote the reaction.

4. Conclusions

A systematic study on the potential active phases including Zn/Cu, ZnO/Cu, and ZnOH/Cu in methanol synthesis by CO_2 hydrogenation is investigated, and the corresponding reaction mechanisms are plot out. It is found that the ZnOH/Cu phase is the most active phase for CO_2 hydrogenation and the following points on the reaction mechanism need to be addressed in the design of catalysts in future:

- During the hydrogenation of carbon dioxide to methanol, carbon dioxide prefers to react with hydrogen to produce HCOO rather than COOH, as HCOO will stand vertically on the Zn bridge sites and tightly bind on the surface. This leads to the more favorable formate pathway comparing to the RWGS + CO-hydro pathway, which is valid on all Zn/Cu, ZnO/Cu, and ZnOH/Cu surfaces.
- Carbon dioxide can dissociate into carbon monoxide and oxygen competing with the direct CO_2 hydrogenation to HCOO. However, the generated oxygen on the surface of copper-zinc alloys is not stable. Zinc oxide will quickly react with the hydrogen or the produced water to form ZnOH/Cu, which has a significant impact on the subsequent reactions. In addition, the hydroxyl group can also be formed in the dissociation of the intermediates, H_2COOH , during the reaction. No matter how the hydroxyl group generates, it will combine with the zinc atom on the copper-zinc alloy surface forming the new stable and active ZnOH/Cu phase.
- When the ZnOH/Cu phase is formed, the reaction rate of methanol synthesis can be significantly increased by promoting the two key steps of HCOO and H_3CO hydrogenation. What should also be noticed is the remaining of some clean Zn/Cu interface is still necessary for CO_2 adsorption, which means the surface should not be fully hydroxylated. That may ask for the proper ratio of CO_2/H_2 mixed gas and the suitable reductive atmosphere in the experimental condition.

CRediT authorship contribution statement

Xiao-Kuan Wu: Methodology, Data curation, Writing - original draft. **Guang-Jie Xia:** Validation, Formal analysis, Supervision, Writing - review & editing. **Zhen Huang:** Investigation, Writing - review & editing. **Deepak Kumar Rai:** Investigation, Writing - review & editing. **Hong Zhao:** Investigation, Writing - review & editing. **Jie Zhang:** Supervision, Project administration, Writing - review & editing, Funding acquisition. **Jimmy Yun:** Investigation, Writing - review & editing. **Yang-Gang Wang:** Conceptualization, Supervision, Methodology, Resources, Writing - review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors gratefully acknowledge the financial support from Guangdong Provincial Key Laboratory of Catalysis (No.

2020B121201002), Southern University of Science and Technology, China (SUSTech) and the computational resource support from the Center for Computational Science and Engineering at SUSTech. X.K.W and J. Z. also thank the financial support by the National Key Research and Development Program of China, China (No. 2017YFC0210905).

Appendix A. Supplementary material

Adsorption energy of intermediates, minor reaction barriers, estimation of rate constant, ZPE and entropy correction, tests on the adsorption energies with dipole correction, the 3D configurations of unit cells, surface energy calculation and the free energies pathway are all provided.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apsusc.2020.146481>.

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